Marco F. Melchor

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CAREER PROFILE

Mechanical Engineering student with field operations experience at Halliburton, thermochemical research at Baylor, and a full-stack software product built from scratch. Focused on solving operational problems with data and engineering analysis.

Simulation and Analysis: MATLAB, LabVIEW **Software: TypeScript / JavaScript, Next.js, Supabase**
Mechanical Design: SolidWorks, AutoCAD **Operating Systems: Windows, macOS**

EDUCATION

BAYLOR UNIVERSITY- Waco, TX **Expected: December 2027**
Bachelor of Science in Mechanical Engineering

- GPA: 3.43
- Minor: Mathematics and Entrepreneurship
- Dean's Academic Honor List: Spring 2024, Spring 2025
- **Relevant Coursework**: Engineering Design I, Machine Design, Thermodynamics II, Conv. & Alt. Energy Systems, Instrumentation & Measurements

EXPERIENCE

Halliburton Energy Services– Brighton, CO & Odessa/Midland, TX **Summers 2025 & 2026**
PE Field Engineer Intern

- Designed and built a valve-actuator parts-tracking and inventory-planning system for frac equipment, giving teams centralized visibility into what was deployed in the field, what was in use, and purchasing and pricing history.
- Developed an RFID-based automation concept for sand logistics to reduce manual errors and improve tracking accuracy.
- Drafted a safety management document to reduce personnel entries into the red zone (high-pressure area at frac sites).

Apollo Phoenix Ventures LLC - Waco, TX **June 2026-Present**
Founder, Apollo Companion

- Founded and developed Apollo Companion, a full-stack vehicle ownership and build platform using TypeScript, Next.js, Supabase, and Vercel.
- Shipped web and mobile features to 80 users, including vehicle management, parts tracking, research, analytics, and a community feed.

Baylor University– Waco, TX **Fall 2025-Spring 2026**
Research Assistant

- Executed high-throughput thermochemical testing for CO/CO_2 cycling, ensuring controlled setups and accurate data collection for dozens of samples daily.
- Modeled and analyzed results in MATLAB to evaluate heat-transfer behavior and assess the feasibility of carbon-based thermal systems for industrial applications.

National Glass Services- Dallas, TX **Fall 2023-Spring 2024**
Project Manager

- Directed installation crews and coordinated with school administrators across 12 school districts, leading crews of up to 25 personnel to deliver safety film projects in compliance with Texas law for security standards.
- Oversaw jobsite workflow, measurements, quality control, and customer communication while independently completing installations across 15,000+ sq. ft.

ADDITIONAL

- Eagle Scout; Assistant Scoutmaster
- President, ECS (Engineering and Computer Science) Dean's Committee (Spring 2026-Present)
- Vice-President, Society of Hispanic Professional Engineers (2025-Present)
- Tau Sigma National Honor Society (Spring 2025-Present)
- Languages: Spanish (Native), English (Fluent)